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Effects of Freemium Strategy in the Mobile App Market: An Empirical Study of Google Play

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ABSTRACT: This paper examines the effect of the freemium strategy on Google Play, an online marketplace for Android mobile applications. By analyzing a large panel data set consisting of 711 ranked mobile apps, we found that the freemium strategy is positively associated with increased sales of the paid mobile apps. Positive trial experience as represented by high review rating of the free version of a mobile app leads to higher sales of its paid version, whereas high visibility of the free version of a mobile app as represented by its product rank does not have a significant impact on the sales of its paid version. This finding suggests that although offering a free trial version is a viable way to improve the visibility of a mobile app, offering a quality free app is more important in boosting sales of the paid app. Moreover, we found

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that the impact of review rating is reduced when the free version is offered, or when the mobile app is a hedonic app, because consumers have the ability to experience the app themselves before purchase. These findings extend understanding of the freemium business model to include a market characterized by simultaneous intra-market competition for both the freemium and paid products and demonstrate how such dynamics may influence sales of the paid products.

KEY WORDS AND PHRASES: Android, freemium, Google Play, mobile apps, online product rank, online reviews.

MOBILE APP REFERS TO APPLICATION SOFTWARE—a type of software that allows the user to perform a specific task that can be installed and run on a range of portable digital devices such as smartphones and tablets. A mobile app can be distinguished from general software by the following characteristics: it is optimized for touch-screen-based mobile devices, it is designed mostly for personal use, it is offered free of charge (except for mobile data use costs) or for sale, and most important, it is made available for download to mobile devices directly through a centralized online marketplace where users can rate and review the app and access various app ranking lists.

Since Apple made a key turning point in the business by launching its App Store in 2008, there has been remarkable growth in the mobile app market. Digi-Capital, an investment bank for mobile apps and games, has forecast that by 2017 mobile apps could reach a revenue of more than $70 billion globally, with nongame apps set to double revenue share from 26 percent to 51 percent [15]. In the United States, as of November 2013, the Android mobile operating system (OS) currently leads the competition with 50.3 percent market share, beating Apple's iOS and Windows Phone, which have 43.1 percent and 4.7 percent of market share, respectively [52]. Furthermore, in terms of percentage of time spent, statistics have shown that mobile apps already have overtaken the mobile web[1] since 2011 [38].

For the Android consumers, Google Play (https://play.google.com) is such a one-stop shop for thousands of Android apps. Initially launched as Google's Android Market (or "the Market"), the Android web portal was rebranded and integrated into one of the categories of Google Play in March 2012.[2] This online storefront has been growing at a rapid pace, surpassing the 500,000 available mobile apps milestone on May 2, 2012 [17], not too long after its record high 10 billion downloads in December 2011 [13] and the previous milestones of 200,000 and 300,000 available apps only a few months before, in April and August 2011, respectively. The Netherlands-based app analytics firm Distimo has attributed the surging numbers to the new "freemium" applications that keep flooding the Market. In fact, two-thirds of Android apps are freemium, establishing the Market as the largest free mobile app store in the world [16].

Freemium, a combination of the words "free" and "premium," is a business model by which a service or a product is offered free of charge, but a premium is charged

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for advanced features, functionality, or related products and services [30, p. 195]. This strategy is widely utilized in the offering of digital-based products or services, including software, games, and web services. In the mobile app market, freemium apps are free to download but typically include offers to upgrade to the paid version, which is ad free or has richer features, to buy additional content or services within the app (i.e., in-app purchase), or to unlock additional levels of game stages.

Today, the freemium strategy is observed in virtually all mobile app markets—including Apple App Store, BlackBerry App World, Google Play, and Windows Marketplace for Mobile. Free apps are often listed side by side with their paid counterparts to provide consumers the opportunity to test run a mobile app before purchase. The flowchart shown in Figure 1 illustrates how a typical consumer selects and purchases a mobile app in a mobile app store such as Google Play.

A consumer may, of course, purchase a paid app directly without trying out its free version. However, a 2013 study reveals that the majority of consumers take advantage of the free apps offering in online mobile app stores [27]. Hence, we believe that the process depicted in Figure 1 represents a typical scenario that can occur when a consumer searches for a mobile app on Google Play. As do other e-commerce sites, Google Play allows consumers to share their feedback on these apps through reviews and ratings. More interestingly, these mobile app stores also publish ranking lists of both free and paid apps based on various factors such as the number of downloads, average rating, and the number of reviews. The simultaneous presence of two competitive markets for both the free and the paid apps and the availability of ranking and rating information in both markets have generated interesting dynamics in the mobile app market.

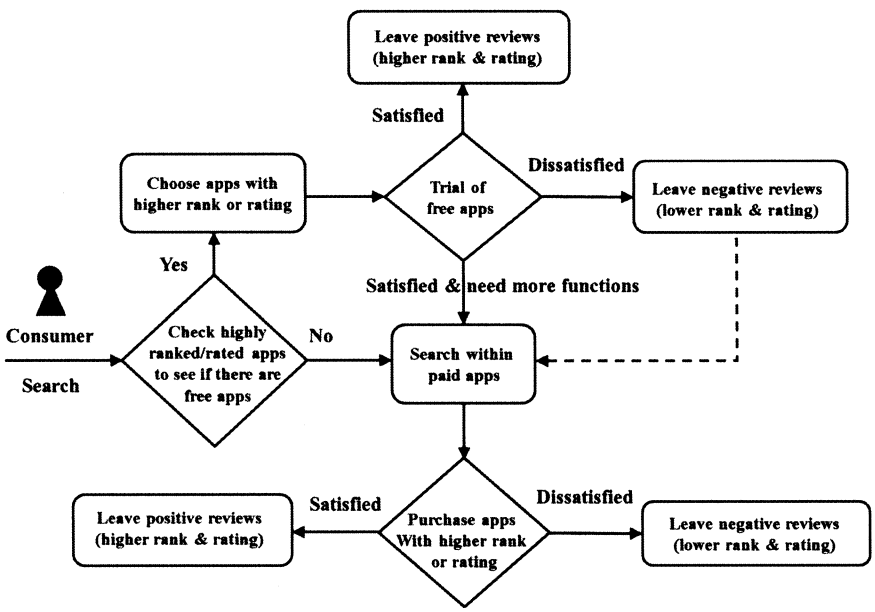

Figure 1. A Typical Consumer Decision-Making Process in the Google Play Online Market

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For example, there are contradictory viewpoints among mobile app vendors about the effect of free apps on their paid version. Some vendors believe that providing free apps decreases the total sales of the paid versions of the apps. Despite the potential revenue streams from free apps' in-app purchases, advertising, and subscriptions, these mobile app vendors still see the traditional pay per download as the most prolific revenue source to target. And to them, the effect of a free app on its paid version is potentially negative, as the free version may cannibalize the sales of the paid version, leading to lower overall revenue and profit.

Other mobile app vendors, however, believe that offering a free version of their app is an effective way to attract consumers, as some consumers who purchased their app would not have done so if they had not experienced the free version of the app. Moreover, a free app that makes it to or near the top of the ranking list can significantly increase the visibility of the app because it will be displayed on the main page of the website. This creates a vicious loop and will lead to more trials of the free app, which in turn can potentially boost sales of the paid version. Hence, these vendors explicitly encourage consumers to try the free version first before making a purchase.

These contradictory viewpoints leave us with several interesting questions: How does the free version of a mobile app affect the sales of its paid version? Do the results learned from use of the freemium strategy in other product markets still hold in the mobile app market? And last but not least, what is the role of the ranking systems for both free and paid apps in the Android mobile app market? To answer these questions, we developed a set of hypotheses and empirically tested them using a longitudinal data set collected from Google Play's website. From our findings, we attempt to offer insights to app developers and marketplace designers into how consumers make their purchase decisions in the presence of widely available free apps.

Literature Review

The Freemium Strategy

The effects of the freemium strategy in the software market have been studied extensively in the literature [10, 22, 28, 41]. Bawa and Shoemaker [6] maintained that free samples can create three potential effects on sales: (1) an acceleration effect, whereby consumers decide to purchase a product earlier than they would have if the free sample had not been given; (2) a cannibalization effect, whereby the free sample reduces sales of the product; and (3) an expansion effect, which occurs when consumers would not otherwise purchase a product without the free sample. Because the outcome of the freemium strategy depends on the interplay of these three effects, which are in turn determined by various factors, the impact of free samples can differ significantly across different categories.

In the traditional software market, in addition to the widely established brand building [37, 41] and learning effects [14], software vendors often exercise the

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freemium strategy to exploit economic rents ascribed to positive network effects [23], product complementarity [47], or an accelerated diffusion process [34]. However, unlike traditional software applications, which often exhibit demand side economy of scale, mobile apps are generally characterized by minimal learning and switching costs and a low degree of interdependence among apps from the same developer. Hence, the classic network effects or product complementarity theory cannot be used to account for the dynamics in the mobile app market. Rather, adoption of paid mobile apps depends critically on two important factors: product visibility and product quality.

Determinants of Success in the App Market: Product Visibility and Product Quality

The importance of product visibility and product quality in determining the success of mobile apps in the market can be explained in the context of a new product adoption process. In the process, consumers generally go through five steps: (1) awareness, (2) interest, (3) evaluation, (4) trial, and (5) adoption [51]. The first step, awareness, is when consumers are informed about the product attributes [36]. In a competitive market environment, as the number of products in a category increases, generating awareness of a new product becomes more difficult [35]. Android app market presents such a challenge not only because there are now close to 1.3 million apps [4] but also because every month some 15,000 new apps are introduced [50]. Product visibility, measured by the rank of the app, is therefore essential to generating a high level of awareness of an app. This is especially true if we consider that most consumers find the apps they are to download through the ranking list published by online marketplaces such as Google Play [1]. Highly ranked apps also generate more interest among potential users because they reflect the interest of other consumers, and such peer influence has been shown to be one of the primary drivers of purchase decisions in online shopping [7].

During a new product's evaluation and trial steps, product quality is critical in determining whether consumers will purchase it. In the software market, consumers are likely to judge the quality of software on the basis of the quality of the free trial version [10]. In the mobile app market, the quality of a free app plays a critical role in the purchase decision of its paid version. This is because many app developers provide free apps with similar quality to their paid version albeit containing ads or with reduced functionality. A study conducted by AdMob [1] revealed that the primary reason for purchasing paid apps was that consumers liked the free version and decided to upgrade to the paid version. Furthermore, other studies have shown that on Google Play and Apple's App Store, a large portion of paid app consumers had tried the free version first before purchasing [49].

In the online environment, product quality can also be inferred from online customer reviews, which are generally left by customers who have purchased and experienced the product, and so the reviews are very relevant and informative. The

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impact of online reviews, along with product ratings and number of reviews, on the purchase decision of customers has been studied extensively in a variety of markets, such as movies [34], online book stores [35], online shopping stores [36], and video games [37]. The results of these studies generally support the idea that online reviews have become a primary reference for consumers, and the consumer-generated information plays a significant role in the purchase decisions of buyers [44]. However, the extent to which online reviews affect consumers' purchase decisions differ due to a variety of factors, such as product type, product visibility, quality and quantity of the reviews, reputation of the reviewers, and freshness of the reviews.

Product characteristics are one of the most significant factors moderating the effect of online reviews. The impact of online reviews on sales of utilitarian products, compared with that on sales of hedonic products, is relatively high due to the focus on product functionality rather than on individual preferences in product evaluations [38]. Similarly, such an impact is more significant in the evaluation of search products [31, 39], whose quality can be evaluated before consumption (e.g., a calculator), than in that of experience products (e.g., wine), whose quality can only be determined after consumption [36, 40].

The popularity of a product may also moderate the impact of online reviews. With thousands of products competing for consumer attention and media coverage, less attractive products receive little attention and product coverage, resulting in a higher degree of product uncertainty. Online reviews thus become a primary source of information for consumers to reduce such uncertainty. Hence, it is not surprising to see that products less popular in the market are more significantly influenced by online reviews [37].

In terms of the direction of feedback, negative online reviews are found to be more critical in the decision-making process than positive online reviews due to the overall risk aversion among consumers [35]. This is consistent with the results of previous research demonstrating that in the traditional brick-and-mortar stores, negative word of mouth more strongly decreases the willingness to purchase than positive word of mouth increases the willingness to purchase [41].

The quality and quantity of reviews also play an important role in increasing sales volume. Prospective consumers tend to rely more on online reviews that are more thoroughly composed [36] and those with a higher number of reviews [34]. In addition, the impact of reviews is more substantial if the reviews are written by reputable users, although such an effect on sales performance decreases over time [42].

Although the foregoing results have been consistently demonstrated in the literature, almost all of the aforementioned studies examined the impact of product visibility and product quality on sales of the same product. Therefore, it will be interesting to see how the free version of a product affects the market performance of its paid version in a market where the freemium strategy is implemented. In particular, it is compelling to ask whether the findings in the extant literature still hold when both the free and the paid versions of the same product are ranked and rated in the same online marketplace.

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Dynamics in Android App Market of Google Play

Google Play is not the first website to publish product ranks and review ratings. With the rapid advance in data analytics technology, more and more electronic commerce sites (such as Amazon.com) have provided product rank and user reviews of the products listed on their websites to assist consumers in making their purchase decisions. In turn, researchers have keenly identified the rich research opportunity associated with the vast amount of data available on these websites and proposed innovative ways to take advantage of the data resources [24, 26]. In the past decade, researchers have studied the relationship between product rank and actual sales and used the demand data recovered from product rank to estimate consumer surplus [7], analyze price elasticity [11], and study product cannibalization [25]. These studies offer a solid theoretical foundation for our research.

Although Google Play adopts a ranking system similar to the ones implemented by many e-commerce sites, one important distinction lies in the simultaneous publication of the ranking lists for both the free and the paid apps on Google Play, which is typically not present in other websites that sell regular commodities. This offers us a unique opportunity to examine the impact of the adoption of the free version of a mobile app on its paid version. On Google Play, the visibility and quality of a mobile app are largely manifested through two important indicators: the product rank and the review rating. In the next section, we develop several hypotheses to investigate how these two factors contribute to the adoption of the paid version of a mobile app.

Research Model and Hypotheses

Impact of a Free App on Sales of Its Paid Version

In the traditional software market, due to the difficulty in obtaining sales data, studies involving the freemium strategy have largely adopted analytic tools, and the findings are highly contingent upon a variety of factors. For example, the performance of sampled software has been found to differ by software categories [22] and strategies for sampling (e.g., full functionality with limited time period or limited functionality without time limitation) [42]. The profitability of the freemium strategy often depends critically on the demand for product compatibility and the strength of network effects [9, 10]. However, even without strong network effects, offering a fully functional software for free in the early stage may boost a software vendor's profit in the long run [34].

In the mobile app market, we expect the net impact of the freemium strategy on sales to be positive due to the unique characteristics of mobile apps. As a highly centralized marketplace, Google Play is an official portal for app developers to publish their apps and is therefore extremely competitive due to the large number of competing apps in each category, the very low search cost, and the ease of comparison across a variety of offerings.[3] This market structure leads to a fourth "competitive" effect beyond the acceleration, cannibalization, and expansion effects

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identified in Bawa and Shoemaker [6]. When a large number of substitutes in a product category are available for free, there is probably a higher degree of cannibalization of these free substitutes than of a paid app's own free version. Moreover, not offering a free version may significantly reduce the awareness of and interest in the product, which may hurt sales more than cannibalization does [3]. Given that cannibalization is more likely to emerge in markets with a high degree of product differentiation [25], and in the mobile app market there exists a relatively low degree of product heterogeneity, one would expect the competitive effect to dominate the cannibalization effect in this market.

In addition to economics reasons, the freemium strategy may also stimulate demand due to consumer psychology. Kim and Morris [40] showed that consumers experiencing a trial of low-involvement products tend to have better affective and cognitive responses to the products than to products requiring high involvement. Similarly, Kempf [39] argued that consumers with personal experience with a product tend to demonstrate stronger belief, confidence, attitudes, and purchase intentions than those with indirect experience with it, such as through advertising. Given the similarity between mobile apps and the product types examined in the foregoing studies, we expect that the freemium strategy would have a similar impact in the mobile app market. Therefore, we predict that offering the free version of a mobile app is more likely to improve the market performance of its paid version, as represented by the number of downloads of the paid app.

Hypothesis 1: Offering the free version of a mobile app is positively associated with a larger number of downloads of its paid version.

Impact of the Rating and the Rank of a Free App on the Rank and the Revenue of Its Paid Version

The existing literature has suggested that product rank and online reviews are good indicators of product visibility and product quality, and both have been shown to have a significant impact on consumer's online purchase decision [12, 19, 45, 53]. In the case of mobile apps, an interesting difference arises from the fact that a lot of mobile apps offer a free trial version for consumers. As mentioned earlier, there is ample anecdotal evidence that shows that consumers interested in purchasing a mobile app always prefer to try the free version when one is available [29, 32]. Consequently, the experience from the free version trial will be highly influential in the subsequent purchase intention. Because Google Play publishes information about the ranks and ratings of both the free and the paid apps, an interesting question arises: How do the visibility and user satisfaction of the free app, as represented by the rank and the rating of its free version, influence sales of its paid version? Drawing on the extant literature, among paid apps that offer a free version, we expect that a higher rank of the free version would increase the visibility of the app, which in turn would result in more trials of the free app. Because some consumers

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who try out the free version of the app will decide to buy the paid version, all else being equal (e.g., the quality of the app), the larger the user base of the free app, the more users will buy the paid version (note that the ratio of conversion might decrease, but the absolute number of users will always increase with more free trials). At the same time, sales of the paid apps also depend critically on the quality of the free app. For a given paid app, the impact of a high-quality free version may be ambiguous. On the one hand, a highly rated free app can effectively promote the sales of its paid version. On the other hand, some users who are very satisfied with the free version will end up not purchasing the paid version, leading to some cannibalization of the paid app. However, if we compare two paid apps that both offer a free version, given the highly competitive nature of the mobile app market and the abundance of substitute products, the app with a lower quality free version is even more likely to discourage potential buyers from purchasing its paid version. In other words, both apps may suffer from the cannibalization effect, but net sales will still favor the app with the better rating. Therefore, for paid apps that offer a higher quality free version (as reflected in the review rating of the free version), we expect the impact of the freemium strategy to be more significant than it is for apps that offer a lower quality free version.

Hypothesis 2a: Among paid mobile apps that offer a free version, the number of downloads of the paid version is positively associated with the rank of its free version.

Hypothesis 2b: Among paid mobile apps that offer a free version, the number of downloads of the paid version is positively associated with the rating of its free version.

Differential Impact of the Review Rating for Apps with and without a Free Version

When consumers search for products that they are not familiar with, they tend to rely on feedback provided by professionals or other consumers who have experienced the product to reduce uncertainty. This is especially true for experience goods such as PC software. However, in the mobile app market, for apps offering a free version, consumers can test the features and functionalities of the apps before purchase, technically turning them into search goods. Consequently, we expect the impact of review rating will be less significant for apps that offer a free version than for those that do not offer a free version. Among mobile apps that do not offer a free version, we expect the findings from the extant literature about experience goods will still hold here. In this case, less popular products typically present a higher risk due to limited professional review coverage and customer feedback. In the absence of this information, it has been shown that the influence of user reviews is more significant in the purchase of less attractive products [53] and increasingly insignificant in the adoption of more popular products [21]. In our context, these findings imply that the

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impact of rating will be more salient for paid apps with a lower rank than for those with a higher rank, which leads to the following hypotheses:

Hypothesis 3a: The impact of the rating of a paid mobile app on the number of downloads is less significant for mobile apps that offer a free version than for those that do not offer a free version.

Hypothesis 3b: Among paid mobile apps that do not offer a free version, the impact of rating on the number of downloads is more significant for paid mobile apps with a lower rank than for those with a higher rank.

Differential Impact of the Rank and the Rating of the Free Version of Hedonic and Nonhedonic Apps

As with traditional software, mobile apps may be categorized into two subgroups: hedonic and nonhedonic. Game and entertainment apps, which are mainly used for fun, can be classified as hedonic apps, and such apps as utility and business programs can be classified as nonhedonic apps. The evaluation of hedonic products is usually subjective and tends to vary from person to person, as it is largely based on individual preferences rather than commonly accepted standards, whereas the evaluation of nonhedonic products is usually objective and consistent across users. Researchers have found that the impact of online reviews on hedonic products is relatively smaller than the impact on nonhedonic products [8]. Such a finding is expected to be more apparent in the mobile app market because there is often a free version of the app. In the mobile app market, because the majority of the free apps offer features and functionalities similar to their paid versions, consumers have plenty of opportunity to thoroughly evaluate the apps before they make a purchase. When it comes to hedonic apps, due to their more engaging nature (e.g., games), consumers tend to spend more time when they get to try the apps, relative to nonhedonic apps. This in turn leads to more confidence in the consumers' own evaluations of the app and, therefore, greater likelihood of the consumers' relying on their personal experience rather than the review ratings of other people to determine whether they will purchase the paid version. Hence, it can be expected that the impact of the rating of the free version of a mobile app on its paid version will be less significant for hedonic apps than for nonhedonic apps.

When consumers rely primarily on their own experience in determining whether to purchase an app, their awareness of the app (and hence the chance the app will be picked for free trial) becomes a more salient factor in ultimately driving the sales of the app. Because there are more than a million apps in the Google Play store, one that makes it to the top-ranked list achieves much greater visibility and hence an increased chance of being purchased.Therefore, we argue that, relative to a non-hedonic app, the chance that a hedonic app will be adopted by a consumer depends more on its visibility to consumers (as represented by its rank[4]) than on the opinion of other consumers (as represented by its rating). Therefore, we expect that the

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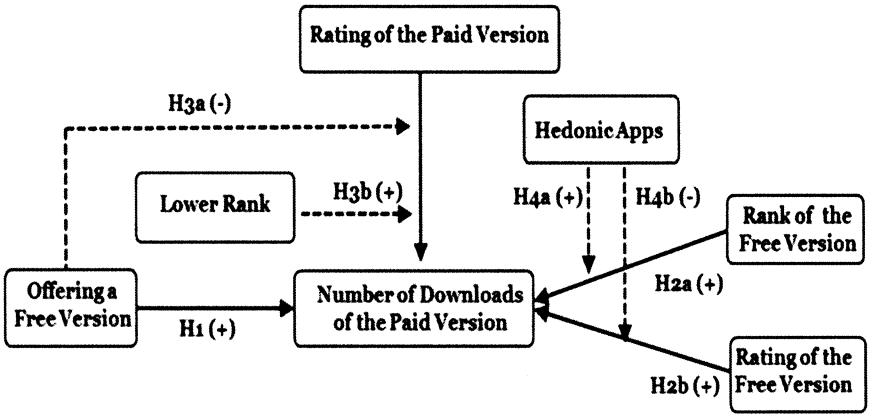

Figure 2. Overall Research Framework

impact of the rank of the free version of a mobile app will be more significant for hedonic apps than for nonhedonic apps.

> *Hypothesis 4a: The impact of the rank of the free version of a mobile app on the number of downloads of its paid version is more significant for hedonic apps than for nonhedonic apps.*

> *Hypothesis 4b: The impact of the rating of the free version of a mobile app on the number of downloads of its paid version is less significant for hedonic apps than for nonhedonic apps.*

To better illustrate the relationships among the various hypotheses proposed in this section, we provide an integrated view of our research framework in Figure 2.

Data and Measures

Sample

To test our hypotheses, we collected data from Google Play and focused on its Android apps store, an online marketplace dedicated to promoting the adoption of mobile apps compatible with the Android platform. In addition to providing links for downloading apps, Google Play's Android apps store also publishes various ranking lists. Among them, the Top Free, Top Paid, and Top Grossing lists are the most relevant for our research questions. The Top Free and Top Paid ranking lists include the top 504 applications in the free and paid categories based on the number of downloads, whereas the Top Grossing list includes the top 504 apps based on their revenue, which consists of both upfront app purchase revenue and in-app purchases (i.e., fees to obtain an advanced feature or to unlock a level in a game). Because these lists are updated daily, we created three software agents to automatically parse on a daily basis the three ranking lists and product web pages related to the ranked apps in each of the three lists.

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We ran the software agents for two months,[5] from November 2011 to January 2012, and collected daily data on the following variables: type of application (paid vs. free), rank of the app in each of the three aforementioned ranking lists, usage category (utility, financial, games, entertainment, etc.), price (for paid apps only), overall user review rating, rating count (number of reviews), developer, app version number, date a new version was released, file size, minimum version of Android OS required, and content rating (level of maturity). During our two-month sample period, 1,567 paid apps, 1,740 free apps, and 704 top-grossing apps appeared on the three ranking lists. [6]

Because we are interested in how the free version of a mobile app affects adoption of the paid version, we used the Top Paid ranking list as the basis for constructing our data set. We created a dummy variable, *FreeVersion,* to indicate whether an application had a free version or not. Of the 1,567 ranked paid applications, 1,020 (65.1 percent) offered a free version on Google Play and were assigned a value of 1 to the dummy variable, *FreeVersion.*[7] Furthermore, among these 1,020 free apps, 379 (37.2 percent) made it to the Top Free ranking list. For these top free apps, we were able to include their Top Free ranks, ratings, and rating counts in our data set.

As the data are both cross-sectional and longitudinal, the initial data set is a large panel data set that consists of 60,142 observations from 1,567 product panels[8] across two months, covering thirty mobile app categories with prices ranging from $0.99 to $39.95. Each product panel represents an Android app i ranked during day t. The app-level panels allow us to control for variation due to the individual characteristics of the app, whereas the time series data allow us to control for variation due to potential "seasonal fluctuations" (e.g., new version being released) and time trends (e.g., declining consumer interest over time).

Variables and Measurement

The data for the majority of the variables were readily available from the web pages we parsed on Google Play. However, a major data challenge in our study was the lack of actual sales data. Like many electronic commerce sites such as Amazon.com and other mobile app providers such as Apple, Nokia, and RIM, due to competitive reasons, Google does not reveal the actual number of downloads for the apps purchased through Google Play. Without these data, we could not test hypotheses related to the impact of the free version of an app on the sales of its paid version. This challenge is very similar to the one confronting many IS researchers who seek to understand the relationship between product rank and actual sales on various online marketplaces [7, 11, 25]. In most of these studies, product rank is publicly available, whereas actual sales data are not. However, it is generally assumed that product rank is largely determined by the actual sales on the majority of these e-commerce sites. Given this assumption, by analyzing the available historical sales and product rank data, researchers have found that there exists an inverse relationship between product rank and actual sales [7, 11, 24, 25], which can often be represented by the power law, or Pareto Type I distribution (Pareto 1896), as shown in the following equation:

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$$Sales = \beta / Rank^{\alpha} + \varepsilon,$$

where β is a scale parameter and α is a shape parameter.

Given this relationship, if the scale and shape parameters are known, researchers can estimate actual sales based on the publicly available product rank. To estimate these two parameters, prior studies have adopted various approaches. Chevalier and Goolsbee [11] examined how the product rank changes in response to a change in demand by purchasing a large quantity of several low-ranked books on Amazon.com. More recently, Garg and Telang [24] developed a simple yet effective approach to estimate the demand parameters by jointly solving the foregoing Pareto equation for both the Top Paid and the Top Grossing ranking lists in Apple's App Store.

Although the foregoing methods have been widely adopted in studies that seek to recover the actual sales from publicly available product rank data, these methods are less applicable to our case. As revealed by several studies [46, 48], unlike other e-retailers, Google Play does not rank its Android apps based solely on the number of downloads. Although the download number still plays an important role in the ranking algorithm adopted by Google Play to unknown degrees, other factors such as rating, freshness of the app, retention rate, and history of growth have all been found to influence the rank [18]. Given the unknowns involved in the ranking algorithm, the inverse function used by prior studies could no longer be applied to our data. In light of this new challenge, we had to resort to another, nontraditional approach to recover the sales data.

After careful investigation of all the information published on Google Play's website, we found that we were able to develop a computational method to recover the download numbers for many, if not all, of the apps in our data set through two pieces of information. The first piece of information was a graph posted on the individual app's web page that shows the fluctuation in the number of downloads for a given app in the previous thirty days. Using graph-to-data digitizing software (i.e., Engauge Digitizer), we were able to recover the relative number of downloads within a given period (i.e., thirty days) for a given app. Table 1 shows sample data resulting from the graph-to-data conversion process.

Table 1. Relative Number of Downloads of a Sample App

Day	1	2	3	4	5	6	7	8	9	10
Number of downloads	40.28	42.63	49.90	52.64	59.53	60.88	64.99	67.72	60.72	55.13
Day	11	12	13	14	15	16	17	18	19	20
Number of downloads	60.64	78.64	91.10	78.54	63.22	60.39	65.90	78.35	83.86	93.53
Day	21	22	23	24	25	26	27	28	29	30
Number of downloads	85.15	74.00	65.61	71.13	76.63	68.24	59.86	55.65	54.21	56.07

Note: The sample data were converted from a Google App's download trend graph showing the fluctuation in the number of downloads of a given app for the previous thirty days.

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However, because Google does not define the vertical axis of the graphs, the converted data from the graphs cannot be used to directly recover the absolute number of downloads. To obtain the actual download counts, we incorporated another piece of information presented on Google Play's app web page, that is, the range of download counts. As of the end of our data collection period (January 18, 2012), Google Play classified the number of downloads for its mobile apps into one of the following eleven categories:

1. 10–50
2. 50–100
3. 100–500
4. 500–1,000
5. 1,000–5,000
6. 5,000–10,000
7. 10,000–50,000
8. 50,000–100,000
9. 100,000–500,000
10. 500,000–1,000,000
11. 1,000,000–5,000,000

On day 5 of our data collection period, the range of download counts changed from the previous day's 500–1,000 to 1,000–5,000, and on day 15 the download range changed from 1,000–5,000 to 5,000–10,000. With this information, we can derive that the number of downloads reached the 1,000 milestone on day 5 and the 5,000 milestone on day 15. Accordingly, we can derive that there were a total of 4,000 downloads between day 5 and day 15. In this case, we can add the relative number of downloads from day 5 through day 15 (which equals 741 based on the data presented in Table 1). We can then divide the total number of downloads (4,000) by the relative number of downloads (741) to obtain the ratio of the actual number of downloads to the relative number of downloads in Table 1. Multiplying this ratio by each day's relative download counts yields the actual number of downloads for this mobile app for the past 30 days.

Following this approach, we were able to recover the actual number of downloads for mobile apps in our data set where there were changes of download ranges during our sample period. However, such changes of download range did not occur for all of the apps in our data set during our sample period. Some apps remained in the same download range, and some only had one change in download range during our sample period. Without the necessary download range change data, we could not recover the actual number of downloads for these apps.

To maximize the number of data points for the number of downloads variable, we checked several other websites to collect additional download range change data to compute the actual number of downloads. For example, we found a website named AppBrain (www.appbrain.com) that provides a comprehensive summary of the activity history of almost all Android mobile apps that have appeared on Google Play. Among

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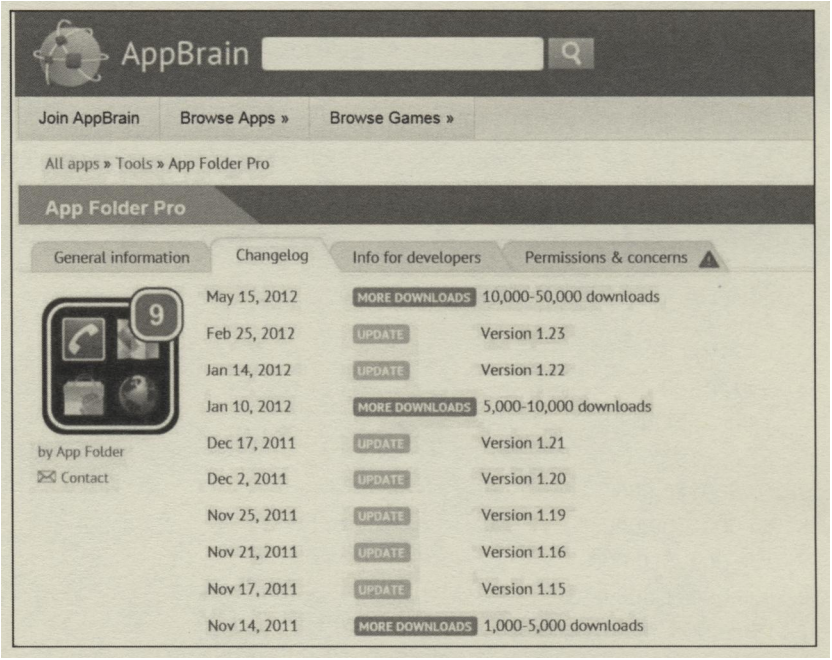

Figure 3. Download Milestone History for a Sample Android App, by AppBrain

the rich information presented on this website, of particular interest to us is the history of the download range change, such as the one shown in Figure 3.

The app highlighted in Figure 3, App Folder Pro, is one of the apps in our data set that had only one change in download range (i.e., on January 10, 2012) during our sample period (from November 18, 2011, to January 18, 2012). But with additional download range change data from the AppBrain website (the number of downloads passed the 1,000 milestone on November 14, 2011), we were able to derive that there were 4,000 downloads between November 14, 2011, and January 10, 2012. Therefore, using the computational method illustrated earlier, we were able to recover the actual number of downloads for this app during our sample period. Note that for the period between January 11, 2012, and January 18, 2012 (the end of our sample period), we can still recover the actual download data for the remainder of the sample period using the download trend graph on January 18, 2012, although the next change in the download range occurred after our sample period (on May 15, 2012), because the actual number of downloads on January 10, 2012, is known.

In addition to AppBrain.com, we also collected download range change data from other sources, such as the developers' websites and from a pilot data set we collected prior to our formal data collection. As a result, we were able to compute and verify the actual downloads for 711 apps out of the total 1,567 apps in our original data set.[9] This resulted in a final data set with 23,771 data points.

Table 2 provides definitions of the key variables and control variables used in our analyses, and Table 3 presents the descriptive statistics.

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Table 2. Definitions of Key Variables

Variable	Definition
$Downloads_{i,t}$	The number of times app *i* has been downloaded (purchased) on date *t*.
$FreeApp_i$	A dummy variable; equals 1 if app *i* has a free version, 0 otherwise.
$RankPaid_{i,t}$	The Top Paid rank of the paid version of app *i* on date *t*.
$RankFree_{i,t}$	The Top Free rank of the free version of app *i* on date *t*.
$RatingPaid_{i,t}$	The user review rating of the paid version of app *i* on date *t*.
$RatingFree_{i,t}$	The user review rating of the free version of app *i* on date *t*.
$Price_{i,t}$	The price of the paid version of app *i* on date *t*.
$Hedonic_i$	A dummy variable; equals 1 if app *i* is considered a hedonic app (i.e., a game or entertainment app)
$DaysReleased_{i,t}$	Number of days since a new paid version of app *i* has been released.
$MinimumOS_i$	The minimum version of Android operating system required to run app *i*.
$FileSize_{i,t}$	The file size of the paid version of app *i* (in megabytes).
$ContentRating_i$	The maturity level of an app, coded with one of the following four values: "everyone," "low maturity," "medium maturity," or "high maturity."

Table 3. Descriptive Statistics of Key Variables

Variable	Mean	Standard deviation	Minimum	Maximum
$Downloads_{i,t}$	265.26	875.06	1	18510
$FreeApp_i$	0.62	0.49	0	1
$RankPaid_{i,t}$	251.92	145.13	1	504
$RankFree_{i,t}$	411.47	289.26	3	504
$RatingPaid_{i,t}$	4.04	0.67	3.71	5
$RatingFree_{i,t}$	4.17	0.52	3.9	4.8
$Price_{i,t}$	2.73	2.54	0.99	39.95
$Hedonic$	0.58	0.49	0	1
$Freshness_i$	66.02	94.51	0	602
$RequiredOS_i$	1.88	0.34	1	4
$FileSize_{i,t}$	9.69	12.10	0.01	99
$ContentRating$	1.51	0.84	1	4

Econometric Models, Estimation Results, and Sensitivity Analysis

Econometric Models

We constructed three econometric models to test the hypotheses proposed earlier under "Research Model and Hypotheses." Given that several hypotheses involve both the main effects and the moderating effects, we created several interaction terms

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and included them in the econometric models along with the key independent variables and a set of control variables.

$$\begin{aligned} Downloads_{i,t} = {} & \alpha_0 + \alpha_1 FreeApp_{i,t-1} + \alpha_2 RatingPaid_{i,t-1} \\ & + \alpha_3 FreeApp_{i,} * RatingPai\,d_{i,t-1} + \alpha_4 Price_{i,t-1} \\ & + \alpha_5 Hedonic_i + \alpha_6 DaysReleased_{i,t-1} + \alpha_7 ContenRating_i \\ & + \alpha_8 MinimumOS_i + \alpha_9 FileSize_{i,t-1} + \varepsilon_{i,t} \end{aligned}$$

Model 1 (for H1 and H3a)

$$\begin{aligned} Downloads_{i,t} = {} & \beta_0 + \beta_1 RankFree_{i,t-1} + \beta_2 RatingFree_{i,t-1} + \beta_3 Price_{i,t-1} \\ & + \beta_4 Hedonic_i + \beta_5 Hedonic_i * RankFree_{i,t-1} \\ & + \beta_6 Hedonic_i * RatingFree_{i,t-1} + \beta_7 DaysReseased_{i,t-1} \\ & + \beta_8 ContentRating_i \\ & + \beta_9 MinimumOS_i + \beta_{10} FileSize_{i,t-1} + \varepsilon_{i,t} \end{aligned}$$

Model 2 (for H2a, H2b, H4a, and H4b)

$$\begin{aligned} Downloads_{i,t} = {} & \delta_0 + \delta_1 RankPaid_{i,t-1} + \delta_2 RatingPaid_{i,t-1} \\ & + \delta_3 RankPaid_{i,t-1} * RankPaid_{i,t-1} + \delta_4 Price_{i,t-1} \\ & + \delta_5 Hedonic_i + \delta_6 DaysReleased_{i,t-1} + \delta_7 ContentRating_i \\ & + \delta_8 MinimumOS_i + \delta_9 FileSize_{i,t-1} + \varepsilon_{i,t} \end{aligned}$$

Model 3 (for H3b)

In all three models, the dependent variable $Downloads_{i,t}$ is the number of downloads of a paid app i on date t. Model 1 is estimated based on all of the apps in our data set and is used to test H1 and H3a, which examine the impact of the freemium strategy on sales and the moderating effect of free apps on the rating of their paid version. Model 2 is estimated against all paid apps that offer a free version and is used to test H2a, H2b, H4a, and H4b, which examine the impact of the rank and the rating of the free apps on their paid version and their differential impact on different app types. Finally, Model 3 is estimated against all paid apps that do not offer a free version and is used to test H3b, which examines the interaction between the rank and the rating of the paid apps.

Because our data set is a typical panel data set, we started the analyses with several specification tests to examine whether our models are subject to specification errors and violations of the ordinary least squares (OLS) estimation assumptions and to determine the appropriate econometric treatments. We started with the standard heteroskedasticity and autocorrelation checks. A Breusch-Pagan/Cook-Weisberg test for heteroskedasticity yields a high χ^2 for all three data sets, suggesting the presence of heteroskedasticity. This is consistent with a plot of the residuals versus fitted (predicted) values, which exhibits a wider scatter with greater x-axis values. The Wooldridge test for autocorrelation in the panel data also shows that first-order autocorrelation (AR1) cannot be ruled out for our data set. This is not surprising, given the longitudinal nature of our data. The presence of

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heteroskedasticity and AR1 autocorrelation suggests against the use of OLS. To address these issues, we applied the generalized least squares (GLS) estimation procedure[10] with corrections to adjust for both heteroskedasticity and panel-specific first-order autocorrelation. We also used the lagged independent variables to account for any potential time lag between the independent variables and the dependent variable. The results reported in Table 4 are coefficient estimates obtained from the GLS estimation.

Estimation Results

To examine the impact of the free version of an app on the sales of its paid counterpart (H1), we regressed the dependent variable *Downloads* on a dummy variable, *FreeApp,* controlling for other factors such as price, rating, app type (hedonic vs. nonhedonic), days since released, file size, content rating, required minimum OS version, and an interaction term for *FreeApp* and *RatingPaid.* With the inclusion of the interaction variable, the main effect of the *FreeApp* variable (H1) in Model 1 is not captured solely by the coefficient for *FreeApp.* Instead, the impact of the interacting variable also needs to be taken into account. More specifically, the main effect of *FreeApp* should be computed as the partial derivative of the dependent variable with respect to the variable of interest. Thus, H1 can be mathematically represented as

$$H1 : \frac{\partial Download_{i,t}}{\partial FreeApp_{i,t-1}} = \alpha_1 + a_3 E(RatingPaid_{i,t-1}) > 0$$

where *E*(*RatingPaid*) is the expected value of the interacting variable *RatingPaid.*

As shown in Table 4, the coefficient for the *FreeApp* dummy (α_1) is positive and significant at the 1 percent level, whereas the coefficient for the interaction term (α_3) is negative and significant>. Substituting the mean value of the *RatingPaid* variable into the above expression shows that the overall value is positive, which suggests that for a paid app with an average rating, offering a free version is associated with more paid app downloads, thus supporting H1.

Similarly, Hypothesis 3a, which predicts that the importance of *RatingPaid* is reduced when a free app is offered, can be mathematically represented as follows:

$$\begin{aligned} H3a : & \left.\frac{\partial Download_{i,t}}{\partial RatingPaid_{i,t-1}}\right|_{FreeApp_i=1} \\ & = \alpha_2 + \alpha_3(FreeApp_i = 1) > \left.\frac{\partial Download_{i,t}}{\partial RatingPaid_{i,t-1}}\right|_{FreeApp_i=0} \\ & = \alpha_2 + \alpha_3(FreeApp_i = 0) \end{aligned}$$

Because the interacting variable *FreeApp* is a dummy variable, the interaction effect can be interpreted by plugging in the two dichotomous values, 1 and 0, into the above inequality. The coefficient estimate for *RatingPaid* (α_2) is positive, but the coefficient estimate for the interaction term (α_3) is negative, indicating that although a paid app's review rating has a positive impact on the number of

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Table 4. Regression Results

Dependent variable: *Downloads*	Model 1 (all apps)	Model 2 (apps with free version)	Model 3[a] (apps without free version)
Constant	−374.98**	−545.34**	−167.15**
FreeApp	238.61**	N/A	N/A
RatingPaid	118.28**	N/A	58.96**
*FreeApp*RatingPaid*	−34.52*	N/A	N/A
RankPaid	N/A	N/A	−0.451**
*RankPaid*RatingPaid*	N/A	N/A	−0.23**
RankFree	N/A	0.41*	N/A
RatingFree	N/A	500.48**	N/A
Hedonic	23.15	25.07**	59.46**
*Hedonic*RankFree*	N/A	−0.50**	N/A
*Hedonic*RatingFree*	N/A	−569.24**	N/A
Price	−5.95*	−6.92**	−1.84
DaysReleased	−0.85**	−0.82**	−0.239**
ContentRating	77.07**	87.95**	61.32**
MinimumOS	26.37	50.63*	114.06**
FileSize	2.56**	2.31**	1.00**
Fit Statistics (Wald χ^2)	1417.18**	1350.02**	2026.43**

[a]In Model 3, continuous variables *RatingPaid* and *RankPaid* are centered to create the interaction term.
*$p < 0.05$; **$p < 0.01$.

downloads, its magnitude is reduced when a free version of the app is offered, thus providing support for H3a.

To test Hypotheses 2a and 2b, we estimated Model 2, which regresses *Downloads* on two new variables, *RankFree* and *RatingFree.*[11] These two variables represent the visibility and quality of the free app. As with Model 1, due to the inclusion of the interaction terms between these two variables and the *Hedonic* variable, the main effects should be computed based on the overall value of the partial derivative of *Downloads* with respect to *RankFree* and *RatingFree,* respectively. The coefficients reported in Table 4 show that both β_2 and β_3 are positive. However, because a higher rank is actually equivalent to a smaller rank number, a positive coefficient on the *RankFree* variable indicates that the higher the rank of a free app, the fewer the downloads of its paid version. Plugging in the mean values of the interacting variables still shows that the overall main effects of these two variables are both positive. Therefore, H2a is rejected and only H2b is supported.

To test Hypotheses 4a and 4b, we examined the coefficient estimates for the interaction terms *Hedonic*RankFree* and *Hedonic*RatingFree.* The coefficient estimate for the interaction term *Hedonic*RankFree* (β_5) is negative. Substituting the values of the *Hedonic* variable (0 and 1) into the partial derivative formula shows

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that when *Hedonic* = 1, the overall impact of *RankFree* is negative (−0.09), which implies that the higher the rank of the free version of a hedonic app, the more the downloads its paid version will enjoy (due to the inverse relationship between ranking position and the actual rank number). When *Hedonic* = 0, the overall effect becomes positive (0.41), which is consistent with the prediction of H4a that the impact of *RankFree* is more important for hedonic than nonhedonic apps. With regard to H4b, the coefficient estimate for *RatingFree* (β_2) is positive and significant, and the coefficient estimate for the interaction term *Hedonic*RatingFree* (β_6) is negative and significant, implying that the impact of the rating of a free app is less significant for hedonic than nonhedonic apps. Hence, H4b is supported.

Finally, to test H3b, we constructed a different data set by including only apps that did not offer a free version. Because the interacting variables *RankPaid* and *RatingPaid* are both continuous variables, in order to avoid multicollinearity and to make the interpretation of the interaction effects more straightforward, we centered the original interacting variables before computing the interaction term [2, pp. 35–36; 33]. The means of the centered variables are zero, and the correlations between the interaction term and the original variables are much smaller. As shown in Table 4, the coefficient estimate for *RatingPaid* is positive and that for the interaction term is negative. Again, since a lower-rank app actually has a higher rank value, this means that as the rank becomes lower (i.e., larger rank value), the importance of rating actually decreases. Hence, H3b is not supported.

Table 5 summarizes the results of our findings.

Sensitivity Analysis

We also performed several sensitivity tests to ensure the robustness of our results. A potential confounding factor in our study is the lag effect of the independent

Table 5. Summary of Findings

Hypothesis	Prediction	Outcome
1	Offering a free app is positively associated with increased sales.	Supported
2a	*RankFree* is positively associated with increased sales.	Not supported
2b	*RatingFree* is positively associated with increased sales.	Supported
3a	The importance of *RatingPaid* is reduced when a free app is offered.	Supported
3b	For apps without a free version, rating is more important for lower-ranked apps than for higher-ranked apps.	Not supported
4a	The importance of *RankFree* increases for hedonic apps.	Supported
4b	The importance of *RatingFree* is reduced for hedonic apps.	Supported

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variables. Although we used lagged independent variables in our estimation, the effects of the product rank and rating of the free version may take longer to realize. To account for this possibility, we took a one-week lag of our dependent variable and reran the regressions. We found that the results are largely consistent with what we reported in Table 4, implying that the lag effect does not play a significant role in our analyses. Moreover, in order to avoid estimation bias due to potential endogeneity that may exist between one of the key independent variables, *RankPaid,* and the dependent variable, *Downloads,* we conducted the standard Durbin–Wu–Hausman test by using the number of tweets as an instrumental variable [24] of the potential endogenous variable *RankPaid.* The number of tweets is an app-specific, self-updated number posted on Google Play that automatically accumulates every time a Google Play visitor clicks the Tweet button on an app's information page. This number shows the number of users who have relayed the app-related information posted on Google Play to their Twitter accounts. This variable is highly correlated with the potential endogenous variable *RankPaid* ($r = 0.619$, $p < 0.01$) because the higher its rank, the more likely an app is to be seen and viewed by consumers, and hence the more likely it will be tweeted about. At the same time, the number of tweets is not directly related to the number of downloads ($r = 0.072$, $p > 0.1$), as relaying the information is not the same as intention to purchase, thus making the number of tweets a good instrumental variable for our endogeneity test. The augmented regression shows that the coefficient for the residual is not significant, thus rejecting the presence of endogeneity in the original model. These tests provided confidence in our results against potential biases caused by the time lag effect and endogeneity.

Last but not least, another potential limitation in our analysis is the fact that Google Play only publishes up to 504 top-ranked apps in each of the free and paid categories. Hence, our sample may be subject to the usual caveats of a truncated sample, especially for Model 1, in which the data set was constructed based on the rank of the paid apps. To see whether our results are biased by the fact that the apps included in our sample are top performers in the Google Play market, we categorized our data into four subsets based on the rank and the review rating of these apps. The threshold values used for the categorization are the top 25 percent, 50 percent, and 75 percent values of these variables. We then reran our models against each subset and compared the coefficient estimates for the independent variables with those reported in Table 4. The results indicate that when changing the rank of the apps, the results are largely consistent with those summarized in Table 4, and the coefficients do not change in a consistent pattern as the rank decreases. However, when estimating the models with different rating groups, the magnitude of key independent variables such as *FreeApp* and *RatingFree* decreases substantially with the rating of the apps, and their coefficients become insignificant when estimated with the lowest rating group. This outcome suggests that our findings may be more applicable to high-quality than to low-quality apps, and a prerequisite for the freemium strategy to be effective is a minimum user satisfaction level with the free app, rather than a high ranking.

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Discussion

The main empirical question in this study is whether offering a free version of a mobile app affects the sales of its paid version, and if so, how the existence of the free version interacts with other product attributes such as review rating and application type (e.g., low-involvement vs. high-involvement product; hedonic vs. nonhedonic product) in shaping a consumer's adoption decision. Our findings provide empirical evidence that helps to answer these questions.

First, we found that offering consumers a free trial version of a mobile app is positively associated with increased sales of its paid version (H1). This finding provides strong support for the freemium strategy adopted by a number of developers in the Android app market. As discussed earlier, the nature of the Android ecosystem has led to a very competitive mobile app market. With thousands of apps available for consumers to choose from, offering a free version is an effective way to increase the awareness and interest of an unknown or newly introduced app. However, some developers are skeptical about the consequences of such a strategy, as they question whether offering a free version of an app might cannibalize sales of the paid version. When a free version of an app is available, consumers who would have paid for the app might choose to continue to use the free version instead of purchasing the paid version. Our empirical findings seem to contradict this outcome and suggest that the benefits of the freemium strategy may outweigh any potential loss. This finding may be attributed to a few possibilities. One, it is likely that cannibalization will still occur to some extent; however, there is strong competition among mobile app developers due to the rapidly growing population of mobile device users and a well-designed centralized marketplace that facilitates ease of comparison. Therefore, a mobile app developer who chooses not to offer a free version may lose more customers to other competitors than it would have lost to cannibalization from its own free trial version. Two, it is also possible that the majority of the consumers who adopt the free version of an app are those who would not have adopted the app if it was offered at a cost, and the free version attracts a lot of customers who are willing to pay for the paid version after they have experienced the free version, leading to an expansion effect. Three, as free apps prevail in the market, consumers will form an expectation that they should be able to try a mobile app for free before purchase, leading to a self-reinforcing feedback loop that further accelerates the widespread offering of free apps in the mobile app market. Regardless of which outcome dominates, it is clear from our empirical results that providing consumers an opportunity to try a mobile app for free actually benefits the developer more than it hurts it.

Having found support for H1, we proceeded to analyze whether the freemium strategy has different impact on different types of mobile apps. Given the similarity in features and functionalities between the free version of most Android apps and their paid versions, it is not surprising to see that rating of the free version positively contributes to the sales of the paid version (H2b).

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However, what is surprising is that the rank of the free version does not have a significant positive impact on the sales of the paid version (H2a), and this finding resonates in our sensitivity analysis, where testing the hypotheses on a subset of the sample with a lower rank does not alter the results. The fact that H2a is not supported can be attributed to several factors. First, as revealed by Google and other market analytics firms,[12] Google's app rank is jointly determined by a variety of factors, such as app title, description, localization, retention rate, number of reviews, app updating, and launch duration, as well as the number of downloads and review ratings. As a result of Google's complicated algorithm, a high rank does not always indicate high user satisfaction or high app quality. Second, mobile app users are more rational, and the commonly observed herd behavior, which suggests that individuals tend to imitate one another [5], does not seem to be as salient in the mobile app market as in other IT markets [21]. Third, availability of the free version of an app in fact reduces the influence of the ranking list, which implies that when consumers are attracted to an app by the high ranking of its free version, they do not automatically take this high rank as a sign of superior quality and automatically purchase the paid version. More than likely, they will still trial the free version to determine if the quality of the app is indeed satisfactory before they make the purchase.

This finding yields important implications for successful implementation of the freemium strategy in the mobile app market. It implies that among apps that offer a free version, those with a higher quality (as represented by the review rating) are more likely to generate a higher level of sales than those with higher visibility but without high product quality. Given that a lot of developers have invested heavily in boosting the rank of their apps in a number of different ways [43], while overlooking efforts to improve app quality, this finding reminds them that the old golden rule of quality determining market performance still holds well in this rapidly evolving market. With the opportunity to try out an app for free, the mobile app market has technically created a virtual fitting room and provides consumers with a shopping experience highly similar to the one found in a traditional department store. Ultimately, consumers will rely more on their own trial experience rather than on product rank or peer review in making their purchase decisions, further stressing the importance of delivering quality products in this burgeoning market.

In addition to analyzing the impact of a free version's rank and rating on sales, we investigated whether such effects were the same for apps with different levels of visibility and for different product types. Contrary to the prevailing notion that review rating has a more significant impact for a less popular product than for a more popular product, our results suggest that when a free version of an app is offered, the importance of review rating diminishes (H3a), which again may be attributed to the fact that consumers have the ability to test the app before purchase, significantly reducing the uncertainty associated with the app, and hence the dependence on user reviews. Moreover, when we narrowed our focus down to apps that do not offer a free version, our prediction that less popular

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apps will benefit more from a high review rating (H3b) is not supported. The rejection of H3b raises interesting concerns about the impact of product rank in the mobile app market. For apps that do not offer a free version, market competition is supposed to resemble what we observe in other electronics markets—that is, most products rely on review ratings to reduce uncertainty in the consumer's decision-making process. In this case, prior research has indicated that the impact of rating will be more significant for lower-ranked products than for higher-ranked products [21, 53]. However, the mobile app market is highly volatile, with thousands of new apps appearing daily and continually refreshing the top of the ranking list. This, coupled with the fact that Google's ranking is not primarily based on app quality or user satisfaction, has resulted in much smaller credit being given to the mobile app ranking list, relative to that in other electronics marketplaces. Hence, it is not surprising that consumers still rely predominantly on app rating, regardless of the ranking of the app.

With regard to the differential impact of the rating of the free version for hedonic and nonhedonic apps, our results suggest that the impact of rating is more significant for nonhedonic than hedonic apps, whereas the impact of product rank is more significant for hedonic than nonhedonic apps. These findings are consistent with those demonstrated in prior studies and imply that consumer psychology observed in online markets for traditional commodities is carried over to the mobile application market. When consumers evaluate a nonhedonic app, they rely more on the evaluation of the quality of the product. However, if the app they consider purchasing is a hedonic product, then their personal experience counts more than peer evaluation.

Our results have important implications for promotion strategies adopted by mobile app developers. In an emerging market characterized with stiff competition and rapid product life cycle, surviving the competition and building a reputable brand can be a daunting task. Other than delivering a solid product that meets consumer's needs, marketing strategy can play an important role in reducing the gap between the market leader and new entrants. Without incurring significant advertising costs, offering a free version of a mobile app can be a cost-effective way to boost sales, but such an approach may only work if the developer offers a high-quality free app. Moreover, our results show that the importance of the rating of the paid version is reduced when consumers have an opportunity to try the product themselves, which is consistent with the fact that consumers rely more on online reviews when there is a high degree of uncertainty, but such uncertainty vanishes with the wide availability of free apps in the mobile app market. Finally, we found that hedonic apps benefit most in sales from offering a free trial, providing clear guidance for mobile game developers and other app developers working in these more personal experience-driven product categories.

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Conclusions and Directions for Future Research

As is generally true with any empirical research, our results are subject to interpretation and are limited to the data available. The most obvious threat to the validity of the conclusions reached here is the longitudinal nature of the data. It is not clear how long it takes for owners of the free apps to upgrade to a paid version. Although we did perform an analysis with a time-lagged model, it may not represent the true pattern of app purchases by free app owners. However, if it takes longer for a free app user to make a purchase decision, then our results are underestimating the positive impact of the freemium strategy. Hence, our results can be interpreted as a conservative, rather than inflated, estimate of this effect. Future studies could complement our research through use of additional research methods such as surveys to obtain a more accurate understanding of the upgrade time frame.

Moreover, although we managed to verify whether an app offers a free version or not by manually checking all the apps offered by the same developer, we were not able to gather data on the number of similar apps in the same product category and thus control for within-category competition. We hope such data may become more easily attainable for future researchers to further understand other competitive factors that may also influence a developer's decision to adopt the freemium strategy.

Last but not least, due to the fact that Google publishes only the top-ranked apps, our sample only includes the top performer in the Android market. Therefore, splitting the sample based on product rank does not allow us to truly differentiate popular apps from less popular apps. Our sample would have been more balanced had we had access to the ranking of the remaining mobile apps.

Despite these limitations, our study is one of the first attempts to empirically assess the impact of the freemium strategy on both sales and revenue of mobile apps. Our findings thus both contribute to the academic literature and provide guidance to practitioners in the mobile app market in several important ways. For researchers, our study builds on the growing body of research that utilizes publicly available e-commerce data to empirically validate research questions. Our research has extended Bawa and Shoemaker's study [5] on the three sampling effects (acceleration, cannibalization, and expansion) by adding a fourth "competitive" effect, accounting for the belief that offering a sample is another crucial element in market competition when there are many companies vying with similar products. In such markets, beyond traditional factors such as price, quality, and marketing efforts, competitive advantage may arise from the adoption of the freemium strategy, as suggested by the higher number of downloads for paid apps that offer a free version. Our study also contributes to the freemium literature by showing that, although improving product visibility through offering a freemium product is helpful, it is more important to offer a quality freemium product, especially for nonhedonic goods. Finally, we have added to the literature on online word of mouth [12, 20, 45] by testing the impact in the mobile app

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market, where both free and paid versions of the same product are available. Not only does our research confirm the findings of previous studies, it also extends our understanding of how consumers' online review (i.e., rating) of a free version of a mobile app can enhance the market performance of the paid version.

Our research also extends existing knowledge of the freemium business model and consumer behavior in online markets to the newly emerged mobile app world. For mobile app market participants, our study provides important guidance for various players in the market. According to a study conducted by Google AdMob in 2009, in making purchase decisions, more than 60 percent of smartphone users rely heavily on the information provided by mobile app online marketplaces such as Apple App Store, Blackberry App World, and Google Play [1]. Building on the results of our work, market designers can facilitate provision of this important information to foster transactions, and developers can prioritize the factors most instrumental in their product promotions. As a result of these improvements, consumers may also benefit from a more efficient, informative, and low-cost shopping experience.

NOTES

1. Mobile web refers to access to the World Wide Web from a handheld mobile device, such as a smartphone, using browser-based Internet services.

2. Google Play also offers thousands of songs, books, and movies for download; but our study focuses only on its mobile apps.

3. According to Android research site AppBrain, approximately 20,000 new apps were rolled out every month in 2012.

4. Note that the rank variable in H4b focuses on capturing the effect of product visibility (the advertising effect), and not on other consumers' downloading behavior (because app rank is not solely dependent on the number of downloads).

5. The mobile app market is a very dynamic and competitive market; hence, we believe that a two-month period is long enough to capture the variation in our key variables without constructing a data set that is too large to analyze.

6. Note that these 704 top-grossing apps may include some of the 1,567 paid apps that made it to the top of the paid ranking list during our sample period.

7. We manually examined all 1,567 of these paid apps to determine whether they offered a free version or not.

8. Note that this is not a balanced panel, due to the fact that not all apps continuously made it to the ranking lists throughout our sample period.

9. Of these 711 apps, 489 were computed using data collected from AppBrain.com, 93 were computed from the download data published on the developers' websites, and 129 were computed using data collected in a pilot data set.

10. We also estimated a fixed effects model. Our main results still hold. However, a fixed effects specification results in excessive collinearity (many of the control variables are dropped in a fixed effects model). Therefore, we adopted the GLS estimator.

11. Note that the data set used to test Model 2 only includes paid apps that offer a free version.

12. www.freemanmobile.com/2013/08/app-store-optimization-improve-ranking-google-play.html.

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